

Informatics Institute of Technology

In Collaboration with

Robert Gordon University, Aberdeen, Scotland

Project Title

Final Year Thesis by

Student Name

RGU ID - Student ID

IIT ID - Student ID

Supervised by

Supervisor Name

April 2026

Submitted in partial fulfilment of the requirements for the
BSc. (Hons) in Artificial Intelligence and Data Science at the Robert Gordon University, Aberdeen,
UK.

DECLARATION

I confirm that the work contained in this BSc project report has been composed solely by myself and has not been accepted in any previous application for a degree. All sources of information have been specifically acknowledged and all verbatim extracts are distinguished by quotation marks.

Date: _____

Student's Name: _____

The above student carried out his research project under my supervision.

Date: _____

Supervisor's Name: _____

**STUDENT PROJECT ETHICAL REVIEW (SPER) FORM**

The aim of the University's *Research Ethics Policy* is to establish and promote good ethical practice in the conduct of academic research. The questionnaire is intended to enable researchers to undertake an initial self-assessment of ethical issues in their research. Ethical conduct is not primarily a matter of following fixed rules; it depends on researchers developing a considered, flexible and thoughtful practice.

The questionnaire aims to engage researchers discursively with the ethical dimensions of their work and potential ethical issues, and the main focus of any subsequent review is not to 'approve' or 'disapprove' of a project but to make sure that this process has taken place.

The *Research Ethics Policy* is available at
www.intranet.rgu.ac.uk/credo/staff/page.cfm?pge=7060

Student Name	
Supervisor	
Project Title	
Course of Study	
School/Department	

Part 1 : Descriptive Questions			
1	Does the research involve, or does information in the research relate to:	Yes	No
	(a) individual human subjects		
	(b) groups (e.g. families, communities, crowds)		
	(c) organisations		
	(d) animals?		
	Please provide further details:		
2	Will the research deal with information which is private or confidential?	Yes	No
	Please provide further details:		

Part 2: The Impact of the Research			
3	In the process of doing the research, is there any potential for harm to be done to, or costs to be imposed on	Yes	No
	(a) research participants?		
	(b) research subjects?		
	(c) you, as the researcher?		
	(d) third parties?		
	Please state what you believe are the implications of the research:		
4	When the research is complete, could negative consequences follow:	Yes	No
	(a) for research subjects		
	(b) or elsewhere?		
	Please state what you believe are the consequences of the research:		

Part 3: Ethical Procedures			
5	Does the research require informed consent or approval from:	Yes	No
	(a) research participants?		
	(b) research subjects		
	(c) external bodies		
	If you answered yes to any of the above, please explain your answer:		
6	Are there reasons why research subjects may need safeguards or protection?	Yes	No
	If you answered yes to the above, please state the reasons and indicate the measures to be		
7	Has PVG membership status been considered?		
	(a) PVG membership is not required.		
	(b) PVG membership is required for working with children.		
	(c) PVG membership is required for working with protected adults.		
	(d) PVG membership is required for working with both children and protected		
	If you answered yes to (b), (c) or (d) above, please give details:		
8	Are specified procedures or safeguards required for recording, management, or storage of data?	Yes	No
	If you answered yes to the above, please outline the likely undertakings:		

Part 4: The Research Relationship			
9	Does the research require you to give or make undertakings to research participants or subjects about the use of data?	Yes	No
If you answered yes to the above, please outline the likely undertakings:			
10	Is the research likely to be affected by the relationship with a sponsor, funder or employer?	Yes	No
If you answered yes to the above, please identify how the research may be affected:			

Part 5: Other Issues			
11	Are there any other ethical issues not covered by this form which you believe you should raise?	Yes	No

Statement by Student			
I believe that the information I have given in this form is correct, and that I have addressed the ethical issues as fully as possible at this stage.			
Signature		Date	

If any ethical issues arise during the course of the research, students should complete a further Student Project Ethical Review (SPER) form.

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Part 6: To be completed by the supervisor			
12	Does the research have potentially negative implications for the University?	Yes	No
If you answered yes to the above, please explain your answer:			
13	Are any potential conflicts of interest likely to arise in the course of the research?	Yes	No
If you answered yes to the above, please identify the potential conflicts:			
14	Are you satisfied that the student has engaged adequately with the ethical implications of the work? [In signifying agreement, supervisors are accepting part of the ethical responsibility for the project]	Yes	No
If you answered no to the above, please identify the potential issues:			
15	Appraisal: Please select one of the following		
	The research project should proceed in its present form – no further action is required		
	The research project requires ethical approval by the School Ethics Review Panel		
	The research project needs to be returned to the student for modification prior to further action		
	The research project requires ethical review by an external body. If this applies please give details		
	Title of External Body providing ethical review		
	Address of External Body		
	Anticipated date when External Body may consider project		

Affirmation by Supervisor			
I have read the student's responses and have discussed ethical issues arising with the student. I can confirm that, to the best of my understanding, the information presented by the student is correct and appropriate to allow an informed judgement on whether further ethical approval is required.			
Signature		Date	

ABSTRACT

Write your abstract here.

PUBLICATION DETAILS

Mention related publications here.

ACKNOWLEDGEMENT

Write your acknowledgements here.

Contents

List of Figures

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Chapter 1

INTRODUCTION

1.1 Chapter Overview

This chapter introduces the research study by outlining its main focus, context, and purpose. It provides a general overview of the topic, explains why it is important, and summarizes the structure of the introduction. The chapter also highlights how the research aligns with existing knowledge and sets the foundation for the following chapters.

1.2 Problem Statement

This section defines the central problem that the research aims to address. It clearly explains the gap or issue identified in the real world or within existing studies that motivates the need for this research. The problem statement should be concise, specific, and evidence-based, showing the reader why the issue is significant and worth investigating.

1.3 Research Motivation

This section explains the underlying reasons for conducting the study. It discusses the practical, theoretical, or personal motivations that led to the selection of the topic. The motivation may include current trends, societal or technological needs, or unresolved challenges found in previous work that inspired this investigation.

1.4 Research Gap

This section identifies the missing elements or limitations in the existing body of knowledge. It critically evaluates prior studies to show what has not yet been explored, adequately addressed, or empirically tested. Highlighting the research gap justifies the originality and necessity of the current study.

1.5 Research Questions

This section presents the key research questions that guide the study. Each question should align with the research problem and define the specific focus areas the study seeks to explore. Well-structured research questions help narrow the scope of the research and form the basis for analysis and discussion.

1.6 Research Aim and Objectives

This section defines the overall aim of the study and lists specific objectives designed to achieve it. The research aim should express the ultimate purpose of the study, while the objectives should break it down into clear, measurable, and achievable steps. These objectives should correspond directly to the research questions.

1.7 Significance of the Research

This section explains the importance and potential impact of the study. It discusses how the findings may contribute to academic knowledge, influence policy, or provide practical solutions. The section may also highlight who will benefit from the research (e.g., practitioners, researchers, organizations, or communities).

1.8 Scope of the Research

This section outlines the boundaries and limitations of the study. It defines the extent of the research in terms of subject area, population, data, tools, and timeframe. The scope helps clarify what is included and excluded, ensuring that the research remains focused and manageable.

Chapter 2

LITERATURE REVIEW

2.1 Chapter Overview

This section provides an overview of the structure and purpose of the literature review. It outlines how previous studies, theories, and models are critically analyzed to build the foundation for the current research. The chapter identifies key themes, gaps, and debates relevant to the research problem, and highlights how the present study contributes to existing knowledge. It also explains the criteria used for selecting and organizing the reviewed literature.

2.2 Background to the Problem

This section introduces the broader context of the research problem. It discusses the theoretical and practical background that led to the identification of the issue under study. Relevant historical developments, key concepts, and domain-specific knowledge are summarized to help the reader understand why this problem is important. The section also establishes the link between the research topic and existing frameworks or theories, setting the stage for the detailed review that follows.

2.3 Theoretical and Conceptual Background

2.4 Review of key methods and Technologies

2.5 Related Work

This section critically examines previous research directly related to the study topic. It reviews existing models, methods, systems, and findings that have addressed similar problems or objectives. Key comparisons are drawn between different approaches, highlighting their strengths, weaknesses, and limitations. The discussion identifies research gaps that the current study aims to address. Where relevant, tables or diagrams may be used to summarize and contrast existing studies for clarity and organization.

The section concludes by synthesizing the reviewed literature to demonstrate how the current research builds upon or diverges from prior work, thereby justifying the need for this study.

Chapter 3

METHODOLOGY

3.1 Chapter Overview

Provide an overview of the chapter and the purpose. Brief the outline of the chapter contents.

3.2 Research Methodology

Identify the research methodology that you follow in your study and give a brief justification on each choice.

3.3 any related sub sections you have

Chapter 4

ANY OTHER CHAPTERS AS REQUIRED

Chapter 5

RESULTS

5.1 Chapter Overview

This chapter presents the findings of the study based on the data collected and analyzed using the methods described in the previous chapter. It provides both quantitative and qualitative results, supported by appropriate figures, tables, and charts where relevant. The purpose of this chapter is to present the results objectively without interpretation, ensuring that the data directly reflects the outcomes of the research questions or hypotheses.

Chapter 6

DISCUSSION

6.1 Chapter Overview

This chapter interprets and explains the significance of the results presented in the previous chapter. It connects the findings to the research questions, objectives, and existing literature. The discussion highlights whether the outcomes support or contradict prior studies and offers possible explanations for any unexpected results. It emphasizes the theoretical, practical, and methodological implications of the findings.

6.2 Any relevant sub sections

Chapter 7

CONCLUSION

7.1 Summary

This section provides a concise summary of the entire research project. It restates the research aim, objectives, and key findings while emphasizing how these findings address the research problem. The summary also reflects on the overall contribution of the study to existing knowledge and practice.

7.2 Limitations

This section discusses the constraints or weaknesses encountered during the research. Limitations may include issues such as sample size, data accessibility, time constraints, methodological boundaries, or assumptions made during analysis. Acknowledging these limitations enhances the transparency and credibility of the study.

7.3 Future Work

This section proposes directions for future research based on the findings and limitations of the current study. It suggests how future studies could build upon this work, overcome identified challenges, or explore new dimensions of the problem. The section encourages continued investigation to expand knowledge and improve applications in the research field.

7.4 Reflective Report

This section provides a personal reflection on the research process, challenges faced, and lessons learned throughout the study. It discusses how the research experience contributed to academic and professional development, problem-solving skills, and subject-matter understanding. The reflection may also include commentary on data collection experiences, time management, ethical considerations, and areas for personal improvement in future research.

APPENDIX